

SMART PET FEEDER

| welcome

Presented by Pp



Pp

Maneerat Sangkaosud 2411311778

Sainam Luo 2411311786

Teekatas Borinayok 2211310020



Project Overview

Smart Pet Feeder is an automatic pet feeding system developed using ESP32. The system can dispense pet food automatically every 10 seconds. Users can also feed their pets manually using a push button.

The system can check the food level and alert the user when the food is low or empty. It can also detect the pet, count the number of times food is dispensed, and count how many times the pet comes to eat after the system starts.

Users can also adjust the food amount to 3 levels: Low, Medium, and High. The system displays food information and system status on an OLED screen, while LEDs show the current system status.



Project Detail



Automatic Feeding

- Dispense food automatically according to the scheduled feeding time.

Manual Feeding

- Allow the user to dispense food manually using a push button.

Food Amount Selection

- Allow the user to select the amount of food : Low, Medium or High.

Food Level Monitoring

- Monitor the amount of food remaining in the container
- Alert the user when the food level is low or empty



Pet Eating Detection

- Detect when the pet comes to feeding area.
- Count an eating time only when the pet stays in the area for 3 seconds.
- The system counts again only after the pet leaves and return.

Status Display

- Display food level, feeding amount, feeding count, and system status on the OLED

Status Indicators

- Use LED to indicate weedy, Feeding and Food Empty status.

Project Specification

Controller

- ESP32

Input Devices

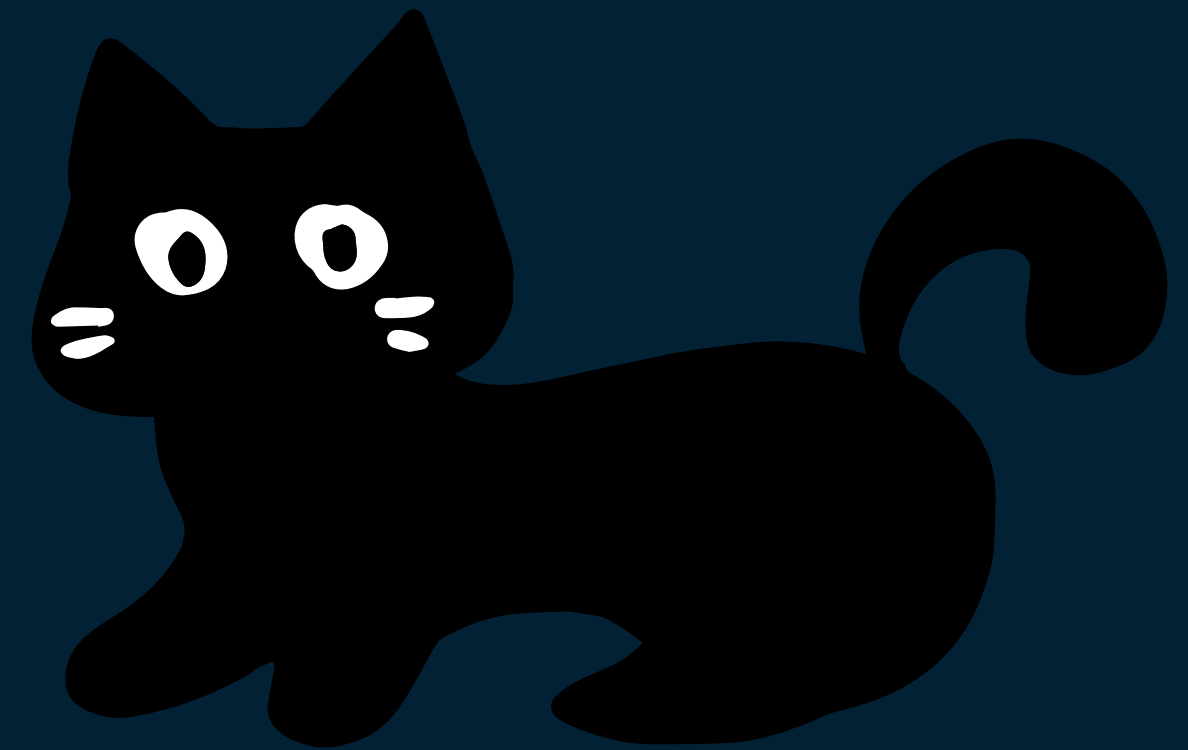
- Push Button
- Potentiometer
- HC-SR04

Output Devices

- Servo Motor
- OLED
- LED

System Parameters

- Food Amount : Low, Medium, High
- Feeding Time : every 10 seconds
- Pet detection Time : 3 seconds
- food Level : 100 % → 0 %
- Communication : I2C for OLED



Project Plan

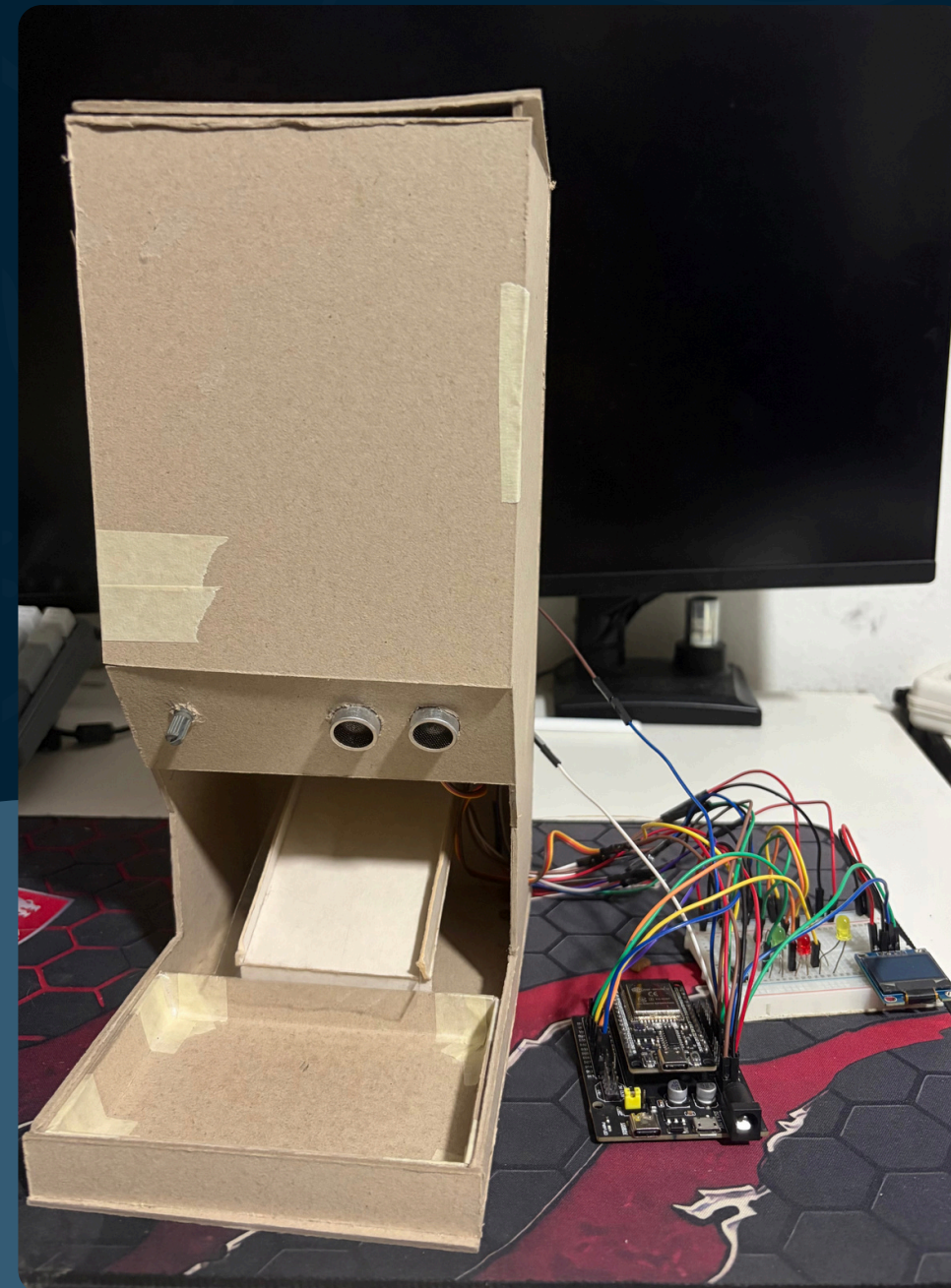
Initial Plan

- 2 LED : Operating / Not Operating
- No pet detection sensor

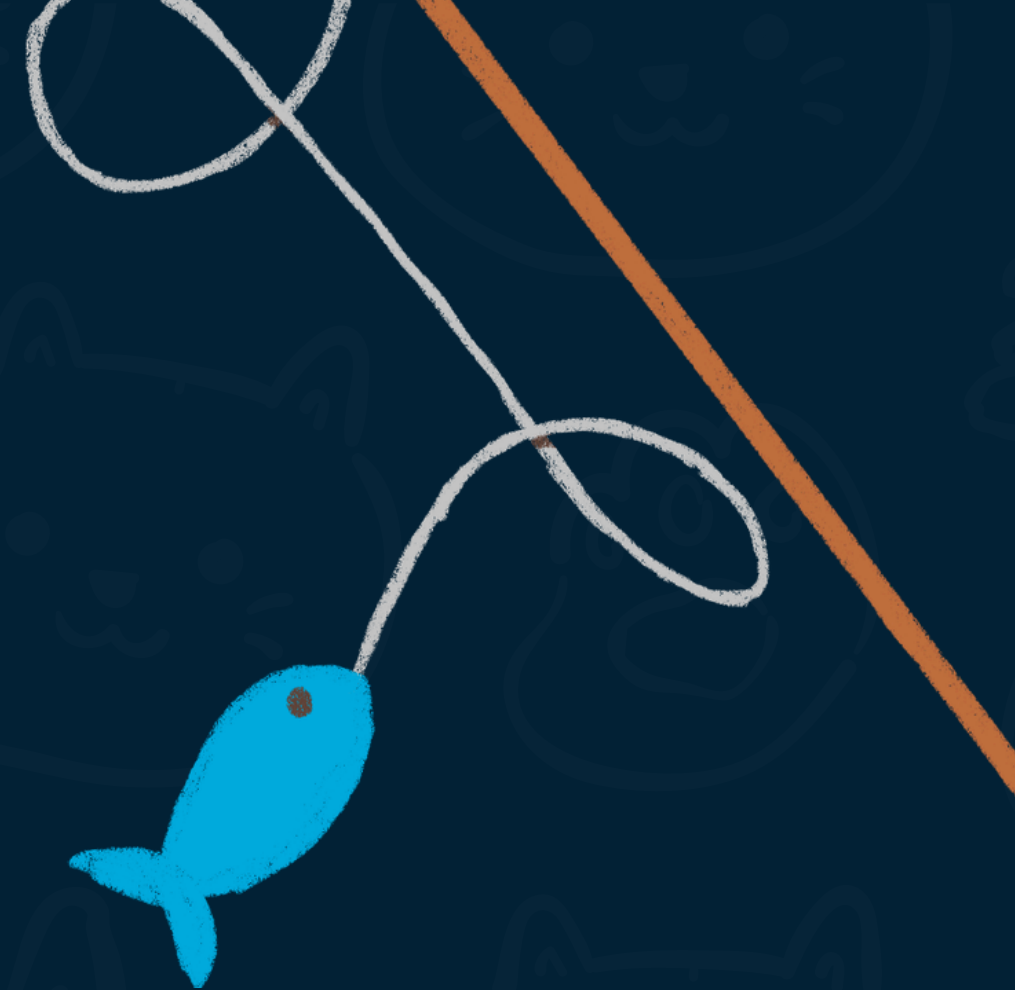
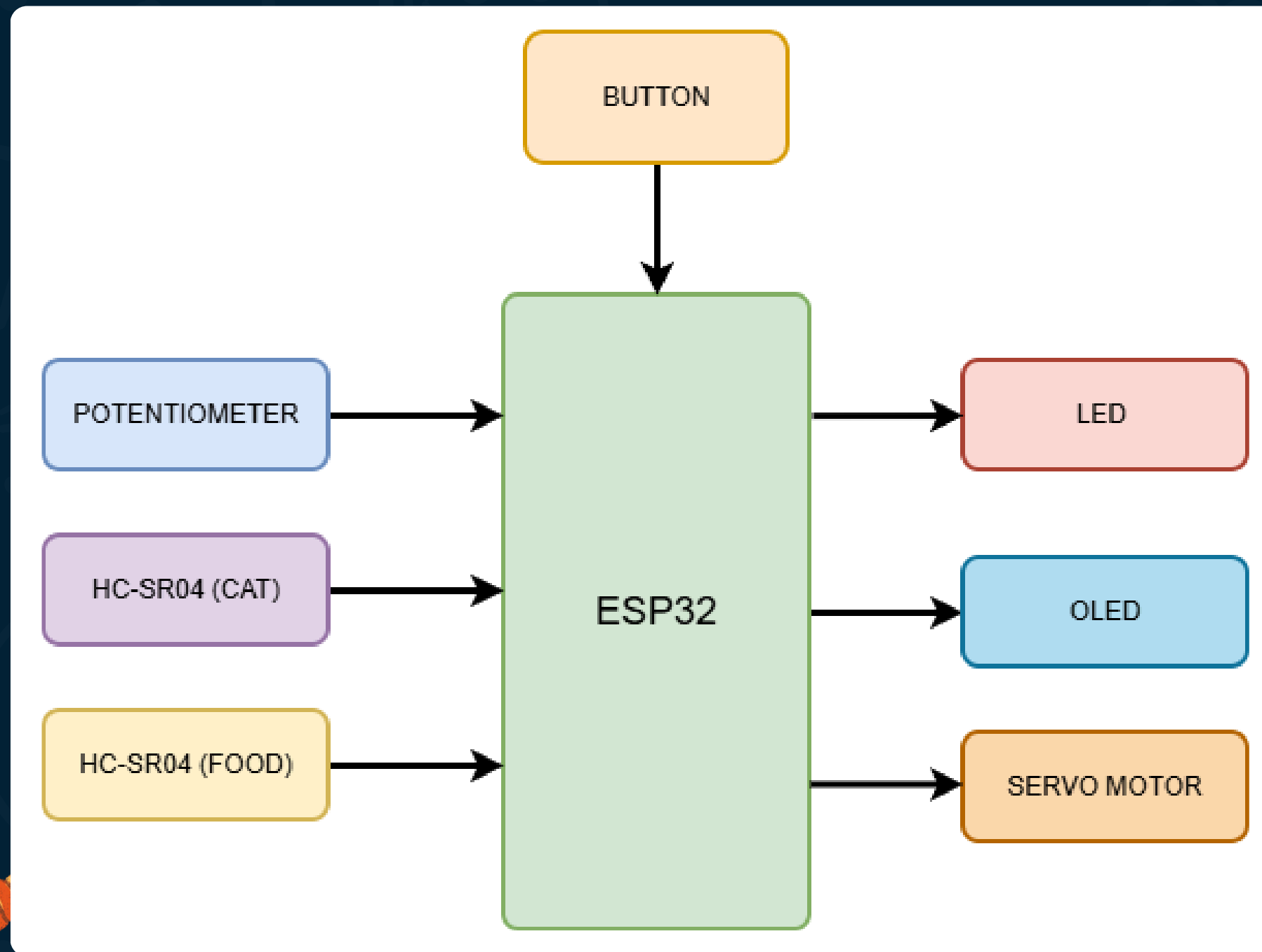


Actual

- 3 LED : Ready, Feeding, Food Empty
- Added pet detection sensor



Architectural design

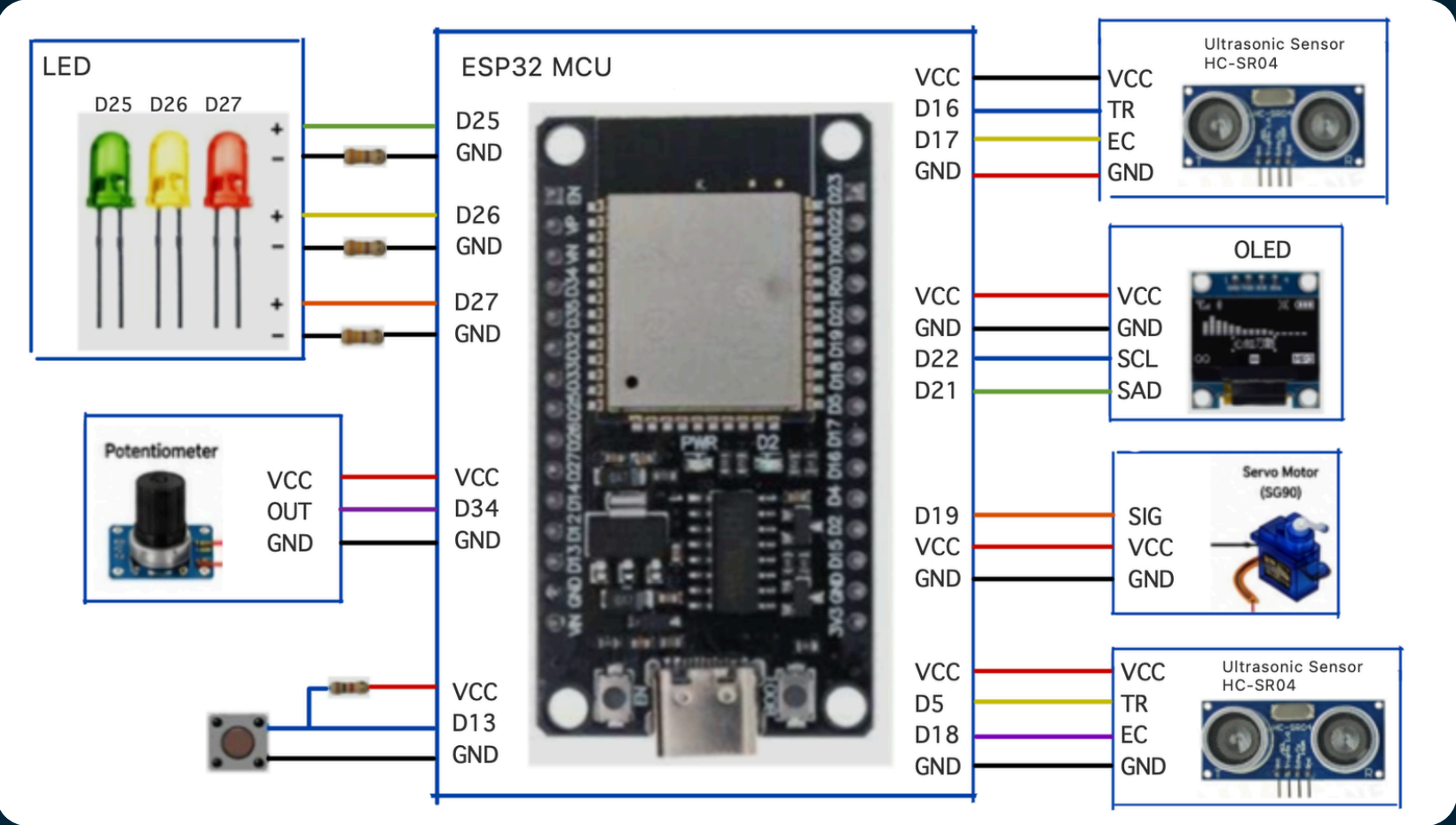


Hardware Selection

ลำดับ	อุปกรณ์	จำนวน	หน้าที่
1	ESP32	1	ควบคุมการทำงานของระบบ
2	เซอร์โวมอเตอร์	1	ควบคุมการจ่ายอาหาร
3	โพเทนชิออมิเตอร์	1	ปรับปริมาณอาหาร
4	อัลตราโซนิกส์	2	ตรวจสอบปริมาณอาหารและตรวจจับสัตว์เลี้ยง
5	OLED	1	แสดงข้อมูลและสถานะการทำงาน
6	ปุ่มกด	1	สั่งจ่ายอาหารด้วยตัวเอง
7	LED	1	แสดงสถานะการทำงาน
8	เบรตบอร์ด	1	สำหรับประกอบและเชื่อมต่อวงจร
9	ตัวต้านทาน	1	จำกัดกระแสในวงจร
10	สายจัมเปอร์	1	เชื่อมต่ออุปกรณ์
11	สาย USB-C	1	เชื่อมต่อกับคอมพิวเตอร์และจ่ายไฟ

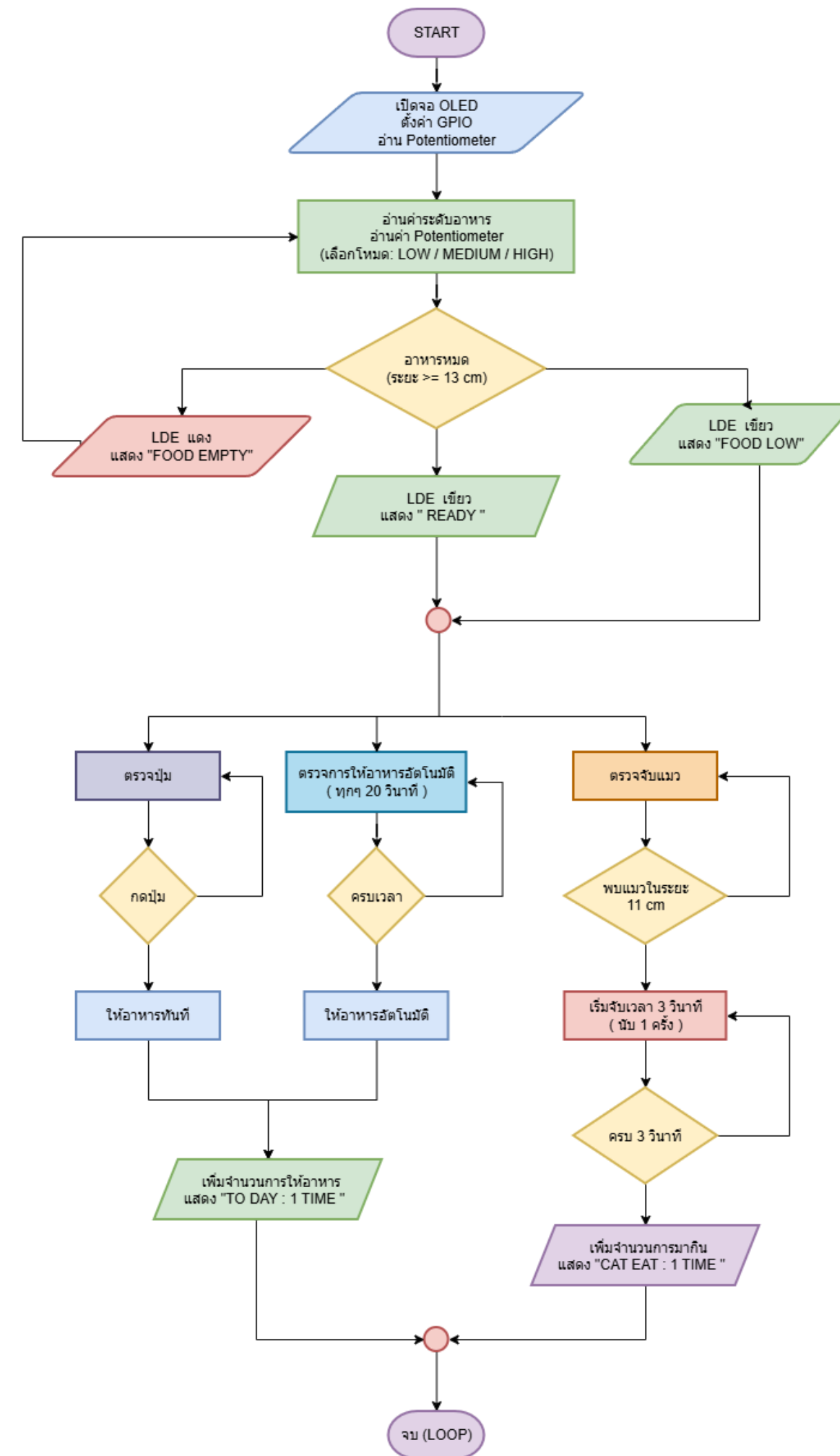


Detail Design



อุปกรณ์	GPIO
Green LED	25
Yellow LED	26
Red LED	27
Push Button	13
Potentiometer	34
Food Ultrasonic TRIG	5
Food Ultrasonic ECHO	18
Cat Ultrasonic TRIG	16
Cat Ultrasonic ECHO	17
Servo	19
OLED SDA	21
OLED SCL	22

Flowchart



Code



3 Levels Food Amount

```
void readAmount()
{
    int adcValue = analogRead(POT_PIN);

    // LOW
    if (adcValue < 1365)
    {
        selectedAmount = "LOW";
        feedTime = LOW_TIME;
    }

    // MEDIUM
    else if (adcValue < 2730)
    {
        selectedAmount = "MEDIUM";
        feedTime = MEDIUM_TIME;
    }

    // HIGH
    else
    {
        selectedAmount = "HIGH";
        feedTime = HIGH_TIME;
    }
}
```

Food Level Measurement

```
bool isFoodEmpty(float distance)
{
    if (
        distance >= FOOD_EMPTY_DISTANCE
    )
    {
        return true;
    }

    return false;
}

// FUNCTION: CHECK FOOD LOW

bool isFoodLow(float distance)
{
    if (
        distance >= FOOD_LOW_DISTANCE &&
        distance < FOOD_EMPTY_DISTANCE
    )
    {
        return true;
    }

    return false;
}
```

Automatic feeding system every 10 seconds

```
void automaticFeeding()
{
    // เก็บเวลาการให้อาหารครั้งล่าสุด
    static unsigned long previousFeedTime = 0;

    unsigned long currentTime =
        millis();

    // ให้อาหารทุก 10 วินาที

    if (
        currentTime - previousFeedTime
        >= 10000
    )
    {
        previousFeedTime =
            currentTime;

        Serial.println();
        Serial.println(
            "AUTO FEEDING"
        );

        Serial.println(
            "EVERY 10 SECONDS"
        );

        feedPet();
    }
}
```



ReadyLED

```
void setReadyLED()
{
    digitalWrite(
        LED_GREEN,
        HIGH
    );

    digitalWrite(
        LED_YELLOW,
        LOW
    );

    digitalWrite(
        LED_RED,
        LOW
    );
}
```

FeedingLED

```
void setFeedingLED()
{
    digitalWrite(
        LED_GREEN,
        LOW
    );

    digitalWrite(
        LED_YELLOW,
        HIGH
    );

    digitalWrite(
        LED_RED,
        LOW
    );
}
```

EmptyLED

```
void setEmptyLED()
{
    digitalWrite(
        LED_GREEN,
        LOW
    );

    digitalWrite(
        LED_YELLOW,
        LOW
    );

    digitalWrite(
        LED_RED,
        HIGH
    );
}
```



Pet Detection System

```
void checkPet()
{
    // อ่านระยะจาก Sensor สัตว์เลี้ยง
    float petDistance =
        readPetDistance();

    // CASE 1 : พบสัตว์เลี้ยง

    if (petDistance <= PET_DETECT_DISTANCE)
    {
        // เพิ่งพบสัตว์เลี้ยง

        if (!petDetected)
        {
            petDetected = true;

            // ยังไม่นับ
            petCounted = false;

            // เริ่มจับเวลา
            petStartTime = millis();

            Serial.println();
            Serial.println("-----");
            Serial.println("PET DETECTED");
            Serial.println("START COUNTING 3 SECONDS");
        }
    }
}
```

```
if (!petCounted)
{
    unsigned long currentTime =
        millis();

    // อยู่ครบ 3 วินาที
    if (
        currentTime - petStartTime
        >= PET_EAT_TIME
    )
    {
        // เพิ่มจำนวนครั้งที่สัตว์เลี้ยงกิน

        petEatCount++;

        // ป้องกันการนับซ้ำ
        petCounted = true;

        Serial.println(
            "PET EAT DETECTED!"
        );

        Serial.print(
            "PET EAT TODAY: "
        );

        Serial.println(
            petEatCount
        );

        // อ่านระดับอาหาร
        float foodDistance =
            readDistance();

        // แสดงผลบน OLED
        showDisplay(
            "PET EAT!",
            foodDistance
        );
    }
}
}
```

```
else if (
    petDistance > PET_LEAVE_DISTANCE
)
{
    // ถ้าเคยพบสัตว์เลี้ยง
    if (petDetected)
    {
        Serial.println(
            "PET LEFT"
        );

        Serial.println(
            "READY FOR NEXT VISIT"
        );

        petDetected = false;

        petCounted = false;

        petStartTime = 0;
    }
}
}
```

Testing and Results

Testing

- Automatic food dispensing test
- Manual feeding using the push button
- Food amount adjustment test (3 levels)
- Food level detection test
- Pet detection test
- OLED and LED status display test

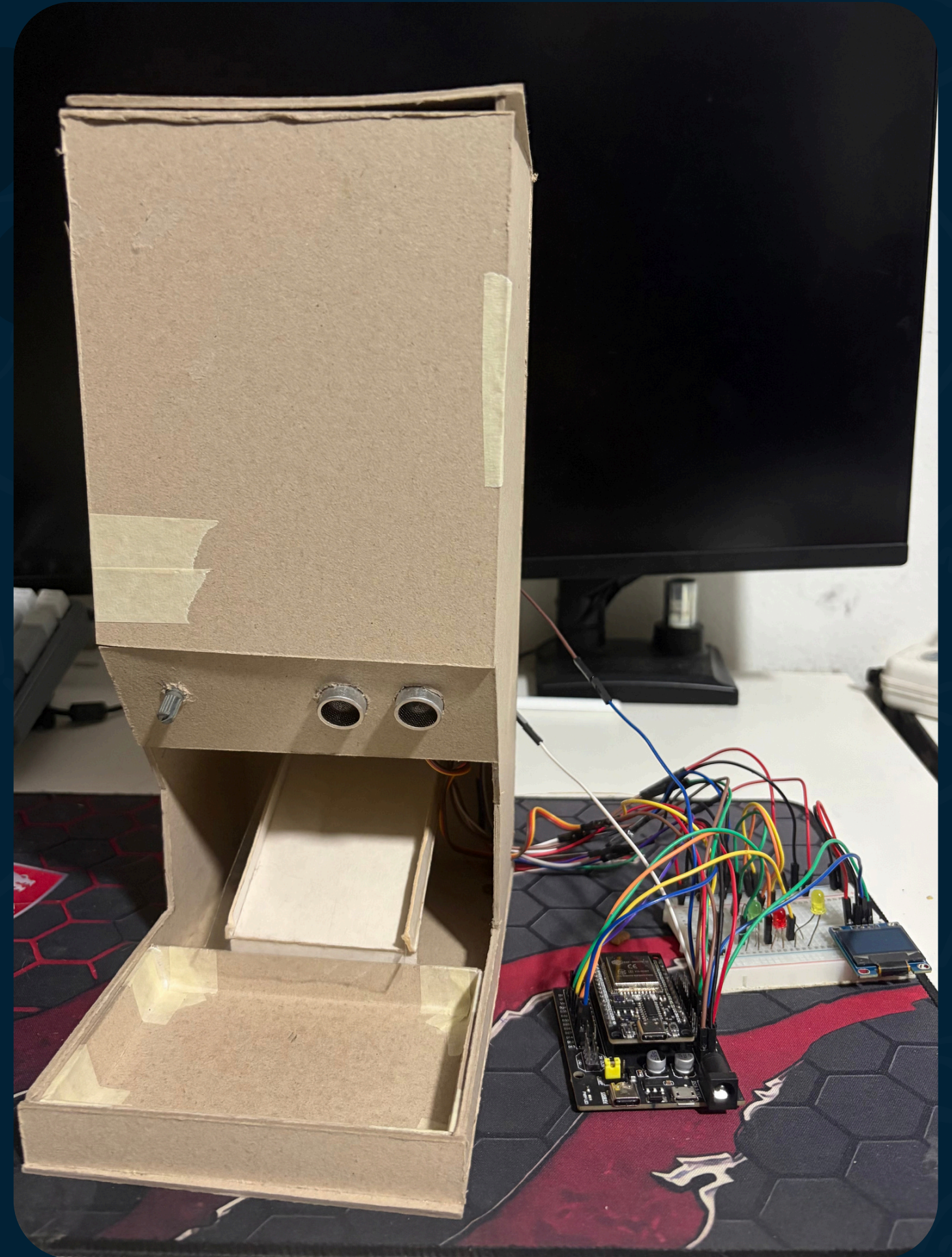
Results

- Food was dispensed successfully.
- Food was dispensed manually using the push button.
- Food amount was adjusted successfully to 3 levels.
- Food level was detected successfully.
- Pet was detected successfully.
- OLED and LED displayed system status



Final

The Smart Pet Feeder can automatically dispense food, adjust the food amount, monitor the food level, detect pets, and display the system status through the OLED and LED indicators.





Link to VDO

Secure toxic substances, cords, plants, and small objects; use gates, ID tags, microchips, leashes, and constant supervision outdoors.



Problems and solutions

1. Ultrasonic Sensor

Problem : The ultrasonic sensor detected the food container instead of the food when placed on the side.

Solution : Moved the ultrasonic sensor to the top of the smart pet feeder so that it could check the food level directly.

2. Servo Motor

Problem : The servo motor worked normally at first, but later it sometimes worked and sometimes did not.

Solution : Changed the circuit and changed the way the food opening was opened and closed.

2. Food Dispensing

Problem : Food got stuck in the dispensing opening, so it could not flow out continuously.

Solution : Widened the dispensing opening to allow the food to flow more smoothly.

Thank you

